

Problem Definition & Relevance

795K

strokes a year in the US · 162,639 deaths (2023)

AHA 2026

1.9M

neurons lost every minute a stroke goes untreated

Saver, Stroke 2006



1 in 8 is a bleed

A clot-busting drug saves a clot and worsens a bleed, so “brain imaging is essential” first.

AHA/ASA 2026

113M

Americans lack 60-min access to clot removal · 49M live 60+ min from advanced care

25%

of patients arrive within 3 hours · scanners live in hospitals

Up to $3M

for one clinical MRI scanner, so brain imaging stays in big hospitals

AHA 2026 · Saver 2006 · AHA/ASA 2026 §3.2 · J NeuroIntervent Surg 2022 · Stroke 2012 · CDC 2025 · NYU Langone 2023

The Team



Anik Sahai (FAU High)

Clinical & product

- Serial Entrepreneur
- Neuroscience Researcher
- Published Researcher in AI

ON THIS PROJECT

Project planning; generated data and implemented algorithms



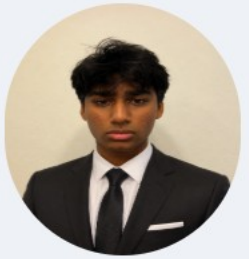
Aaron Loh (American Heritage)

Math & numerics

- FAU: PDE surrogates, proofs
- 1st nationally, Matrices & Vectors
- 2nd, Florida State Science Fair

ON THIS PROJECT

Analyzed algorithms for classical image reconstruction



Charan Jarugula (FAU High)

Algorithm Development

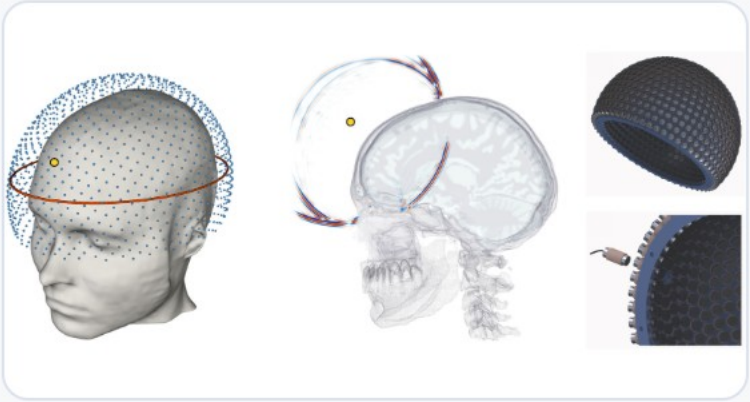
- FAU quantum research: error mitigation
- Computational neuroscience, UF Scripps

ON THIS PROJECT

Analyzed quantum to optimize current classical methods

Problem Definition & Relevance

How ultrasound screening of the brain works: a helmet of speakers and microphones.



WHY NOT A NORMAL ULTRASOUND?

Hospital ultrasound bounces high-frequency sound off tissue and reads the echo. The skull blocks and scrambles that, so it can't see the brain. The helmet uses lower frequencies that pass through bone, and listens from every side.

1

Fire

One transducer sends a short pulse of sound into the head.

400 kHz peak · 3-cycle burst

2

Listen

Every other transducer records the pressure that reaches it, through skull and brain.

240 µs per recording

3

Repeat

Each of the 1,024 transducers takes a turn as the speaker.

>1 million recordings

No radiation, no magnet, small enough to travel. The catch: the recordings are not a picture yet.

Guasch et al., npj Digital Medicine 3, 28 (2020): Methods and Fig. 1, CC BY 4.0

Quantum Rationale & Understanding of Potential

The science: a brain image is a map of how fast sound travels.

1

Voxels

$c = (c_1, \ldots, c_N)$

~10^8 cubes of 0.5 mm, each one sound speed.

2

Travel time

$t \approx \Sigma\ \ell_i / c_i$

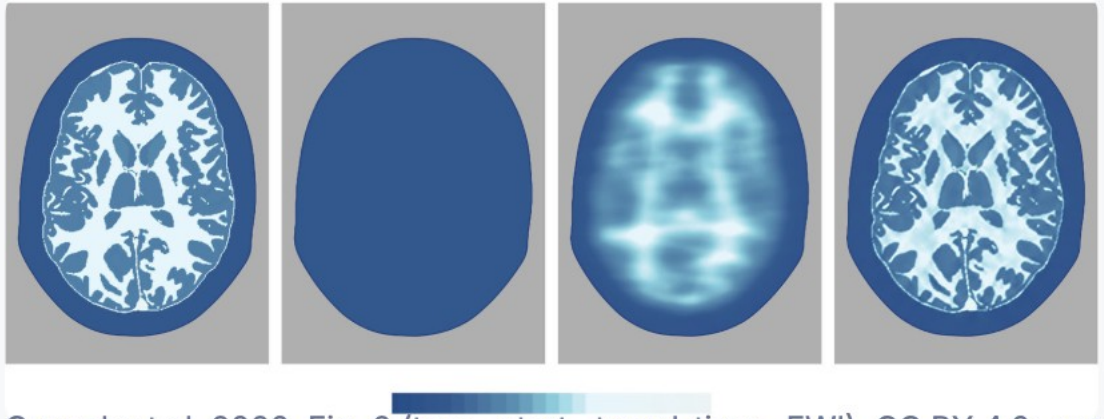
Crossing paths triangulate each cube.

3

Full waveform

$19\ \text{mm} \rightarrow 0.9\ \text{mm}$

Using the whole wave sharpens the image 20×.

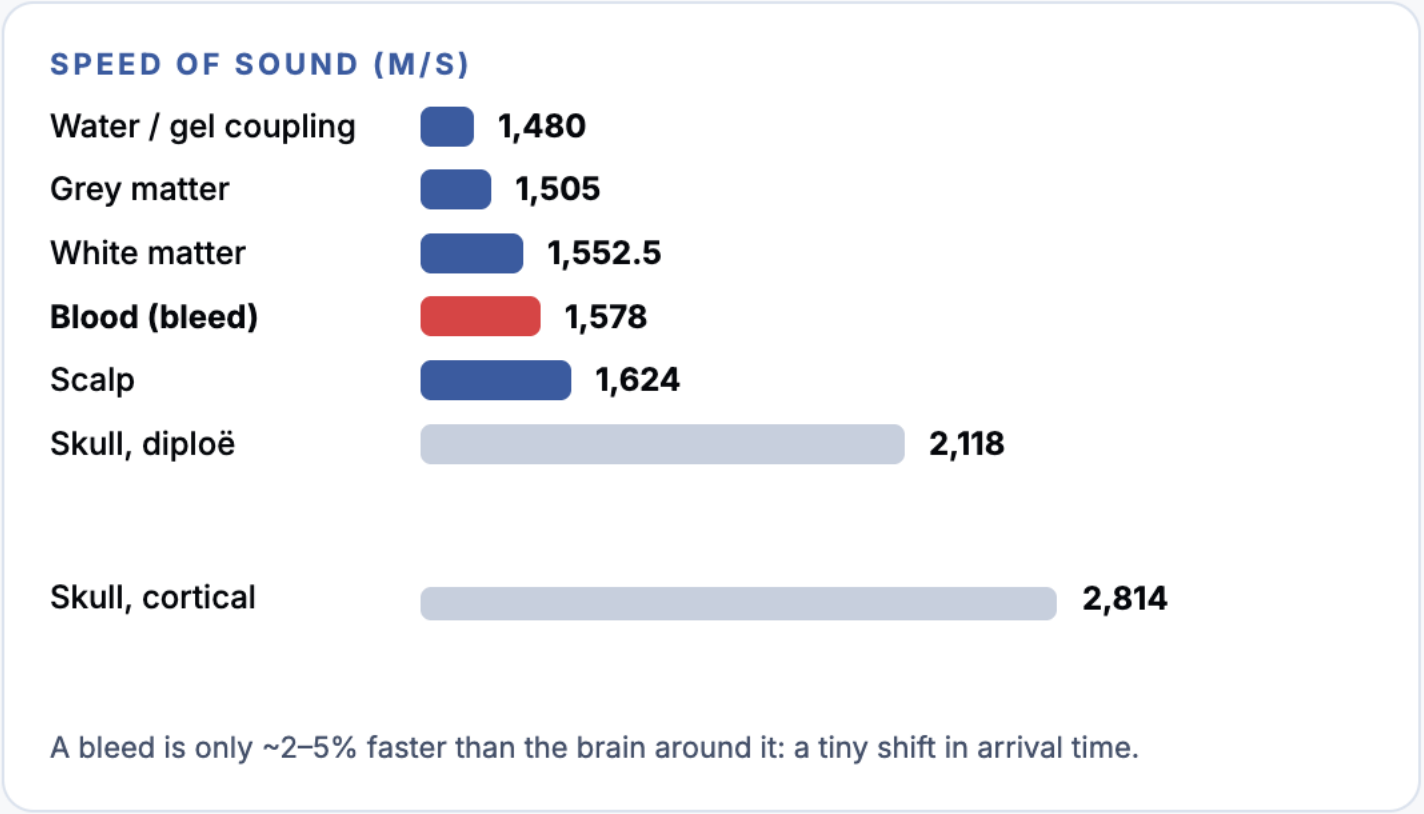


Guasch et al. 2020, Fig. 2 (true · start · travel-time · FWI), CC BY 4.0; supplement p.5

That sharpness is exactly what makes the computation so expensive.

Problem Definition & Relevance

What the recordings mean: every tissue carries sound at its own speed.



A bleed is only ~2–5% faster than the brain around it: a tiny shift in arrival time.

Arrival times are the clue

Sound speeds up in skull and slows in brain. Each path through the head arrives a little early or late, and a bleed shifts the timing on every path that crosses it.

The whole wave, not just the echo

Full-waveform inversion uses the entire recorded wave to rebuild a 3D map of sound speed at 0.5 mm detail. That map is the brain image.

Turning a million recordings into that map is the step that takes ~32 hours.

Tissue sound speeds: Guasch et al. 2020 supplementary table · 0.5 mm grid: Methods

Problem Definition & Relevance

Our baseline, from the paper itself: one brain image takes ~32 hours.



npj | Digital Medicine

www.nature.com/npjdigitalmed

Check for updates

ARTICLE OPEN

Full-waveform inversion imaging of the human brain

Lluís Guasch[1✉], Oscar Calderón Agudo[1], Meng-Xing Tang[2], Parashkev Nachev[3] and Michael Warner[1]

Magnetic resonance imaging and X-ray computed tomography provide the two principal methods available for imaging the brain at high spatial resolution, but these methods are not easily portable and cannot be applied safely to all patients. Ultrasound imaging is portable and universally safe, but existing modalities cannot image usefully inside the adult human skull. We use in silico simulations to demonstrate that full-waveform inversion, a computational technique originally developed in geophysics, is able to generate accurate three-dimensional images of the brain with sub-millimetre resolution. This approach overcomes the familiar problems of conventional ultrasound neuroimaging by using the following: transcranial ultrasound that is not obscured by strong reflections from the skull, low frequencies that are readily transmitted with good signal-to-noise ratio, an accurate wave equation that properly accounts for the physics of wave propagation, and adaptive waveform inversion that is able to create an accurate model of the skull that then compensates properly for wavefront distortion. Laboratory ultrasound data, using ex vivo human skulls and in vivo transcranial signals, demonstrate that our computational experiments mimic the penetration and signal-to-noise ratios expected in clinical applications. This form of non-invasive neuroimaging has the potential for the rapid diagnosis of stroke and head trauma, and for the provision of routine monitoring of a wide range of neurological conditions.

npj Digital Medicine (2020)3:28 ; https://doi.org/10.1038/s41746-020-0240-8

INTRODUCTION

No universally applicable means of imaging the living human brain at high anatomical resolution exists. The modality with the best spatial resolution and tissue contrast, magnetic resonance imaging (MRI), is contraindicated where the presence of magnetic foreign bodies cannot be excluded, and is impractical with claustrophobic, uncooperative, or severely obese patients. Its nearest rival, X-ray computed tomography (CT), involves exposure to harmful ionising radiation. Both require large, expensive, immobile, high-power instruments that are near-impossible to deploy outside specialised environments. The clinical consequences of this are high symptom-to-image times, long inter-scan intervals during serial imaging, and constraints on the range of patients that can be imaged successfully.

portable three-dimensional (3D) quantitative imaging that is less-expensive, faster, and more easily applicable than MRI, and that is safer and has better soft-tissue contrast than CT. This approach results in a three-dimensional, sub-millimetre-resolution, quantitative model of acoustic wave speed within the brain and surrounding tissue, which is capable of distinguishing most of the structures and pathologies to which MRI is sensitive. The combined findings of our in silico and laboratory experiments demonstrate that recording transcranial ultrasound data are feasible, recorded signal-to-noise levels are high, and the data contain information that is sufficient both to build an accurate model of the skull and to reconstruct brain properties. To achieve this, we use two closely related techniques: adaptive waveform inversion (AWI)[3] to build an accurate skull model, and FWI[4] to extract brain properties at high resolution.

~32 h per image

on **128 CPU nodes × 24 cores** (3,072 cores)

> The computational effort required for 3D FWI is considerable. The results shown in Fig. 3 required ~32 hours elapsed time to complete, running on a conventional cluster of 128, CPU-based, 24-core, compute nodes. Our target is to reduce this elapsed time to below 10 minutes; this requires a speed up of ~200 times. The

Their own target: under 10 minutes, a ~200× speed-up. That gap is what we attack.

Classical

REPEAT

- Guess the speed of sound at every location
- Emit a pulse from one transmitter
- Simulate pressure propagation through the model
- Predict pressure at each sensor
- Compare predicted and observed signals
- Repeat for other transmitters
- Gradient descent: update the speed estimate
- Speed of sound everywhere
- Predict tissue arrangement

Sound-wave propagation: tissue changes the wave

Real 2D wave simulation



Speed of sound (m/s)

Water	1,480
Scalp	1,624
Skull, cortical	2,814
Skull, diploë	2,118
CSF	1,505
Grey matter	1,505
White matter	1,553
Blood (bleed)	1,578

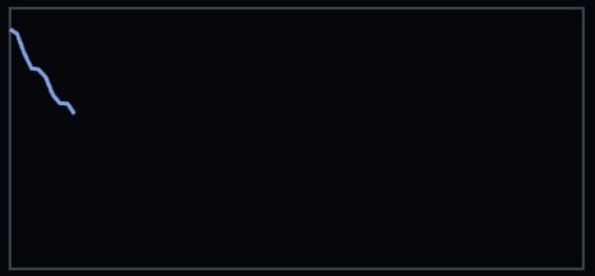
Wave is in: Grey matter · 1,505 m/s

Improving the tissue map with gradient descent

Sound-speed map



Mismatch E



Round 9 / 80

Why Quantum?

Accelerate the wave solver that reconstruction calls again and again.

>95%

of FWI compute is spent propagating waves.

A faster solver could reduce reconstruction time and compute cost.

Why quantum fits

Map waves to quantum evolution; compact encoding may improve scaling for selected measurements.

Near term: validate the method

Small, noisy circuits test accuracy.

Future: larger 3D models

Scaling beyond our demo requires more encoded locations, longer simulations, and deeper circuits. Must beat classical software at similar cost and accuracy.

Our target: quantum wave propagation inside a classical reconstruction loop.

Guasch et al. 2020 supplement

Quantum Rationale & Understanding of Potential

Our literature review: 132 sources across stroke medicine, ultrasound imaging and quantum computing.



Guasch 2020

The helmet and FWI · npj Digit Med



Cueto 2022

Stride: their solver



Costa 2019

Wave equation on qubits



Schade 2024

1D wave on IBM hardware



Bösch 2025

Sources, loss, real speed-up



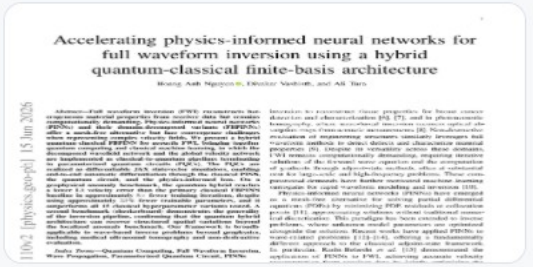
Babbush 2023

When speed-ups are exponential



Harrow 2009

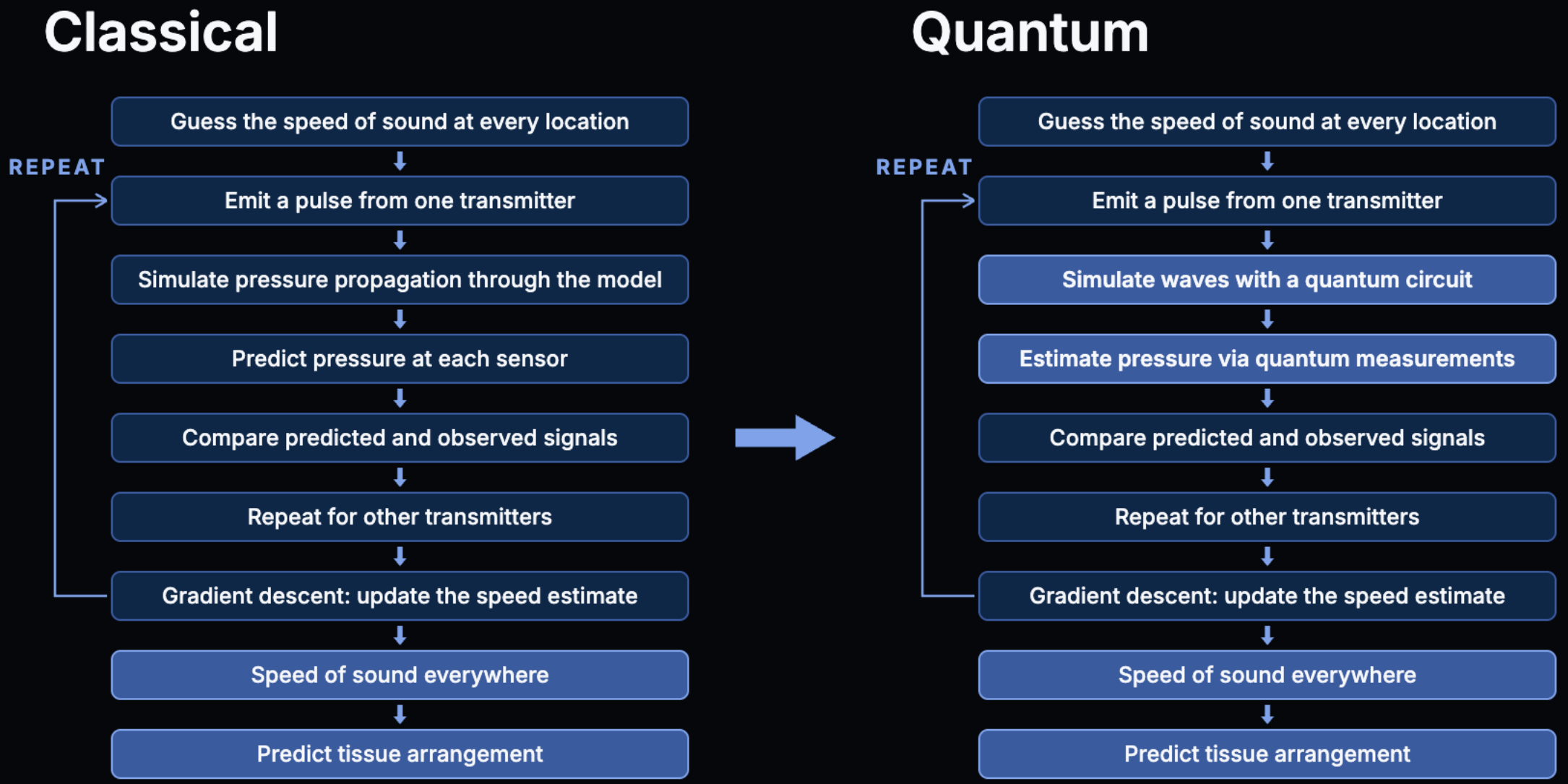
HHL and its readout limits



Nguyen 2026

Hybrid quantum-classical FWI

Four research reports: clinical statistics · ultrasound and FWI · quantum algorithms · audit of our own code. Every number in this deck links to a source.


Classical
REPEAT
Guess the speed of sound at every location
Emit a pulse from one transmitter
Simulate pressure propagation through the model
Predict pressure at each sensor
Compare predicted and observed signals
Repeat for other transmitters
Gradient descent: update the speed estimate
Speed of sound everywhere
Predict tissue arrangement
Quantum
REPEAT
Guess the speed of sound at every location
Emit a pulse from one transmitter
Simulate waves with a quantum circuit
Estimate pressure via quantum measurements
Compare predicted and observed signals
Repeat for other transmitters
Gradient descent: update the speed estimate
Speed of sound everywhere
Predict tissue arrangement

Gate-based quantum algorithm (1/2)

Given a tissue map and a pulse, predict where the wave energy goes.
The larger imaging loop adjusts the tissue map to match real sensor recordings.

01

GUESS TISSUE MAP

Assign sound speeds to the cells. Start a known ultrasound pulse.

02

STORE WAVE IN QUBITS

Quantum gates move the simulated wave between neighbouring sites.

03

ENERGY AT BRAIN SITE

Repeat the circuit. Count how often the selected brain site is measured.

Hardware demo: receiver energy. Pressure reconstruction uses a separate compact circuit.

Gate based quantum algorithm (2/2)

First, connect separate pairs



Then, shift the connections



Demo: 15 qubits · 8 time steps. Diagram shows six encoded positions.

One line is a coupling operation

It lets the simulated wave move between two connected positions.

Tissue speeds set each gate's strength

Different tissues change how the simulated wave travels.

Switch the pairs and repeat

The wave can now pass beyond its original pair of positions.

Simulated Quantum Annealing: Method

01 DEFINE THE STEP

Write the next wave update as a set of linear equations.

02 ENCODE IN BITS

Represent signed updates with bits. A QUBO score penalizes equation errors.

03 SEARCH

CPU-based SQA seeks low-score bit strings while simulated quantum fluctuations decrease.

04 DECODE & REPEAT

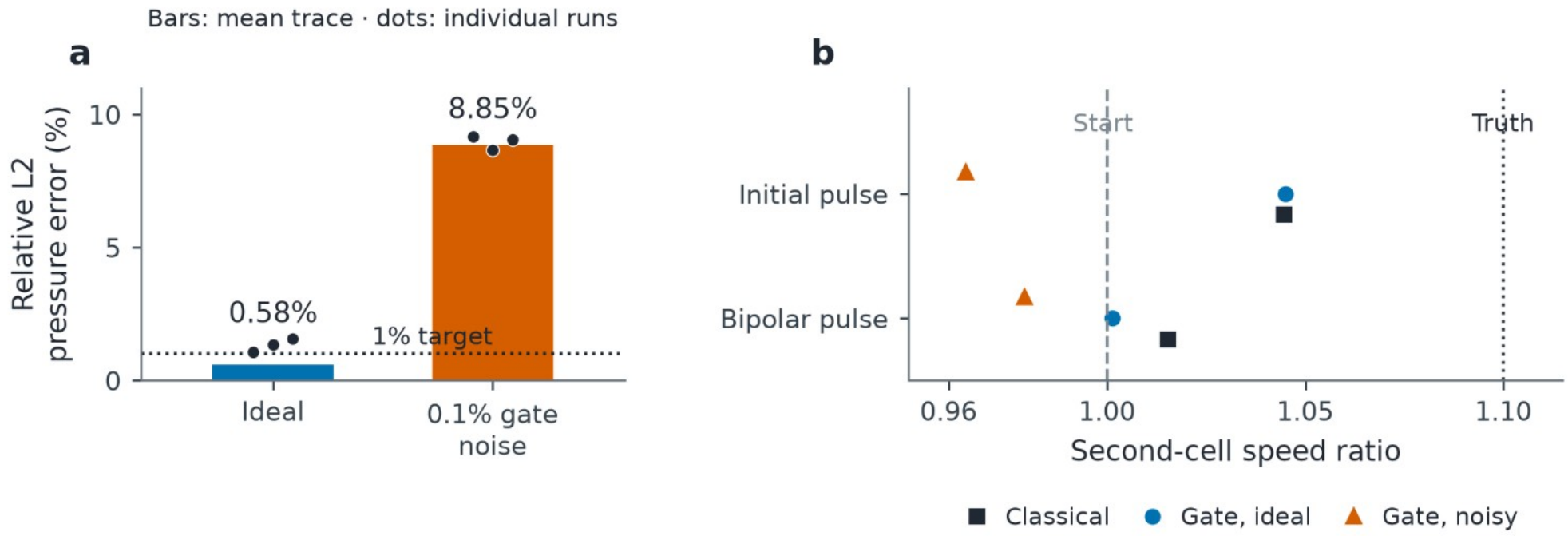
Decode and refine the update; advance time. Repeat to predict the pressure recordings.

	Gate-based method	Annealing method
How it advances	Gates evolve the encoded wave	Search solves each wave update
How values are stored	Quantum amplitudes	Signed bits; default 6 per component
Our test platform	Simulator + IBM gate hardware	Classical CPU; no annealing hardware

Implementation: annealing/forward.py + qubo.py · Crank–Nicolson → binary least squares → path-integral SQA.

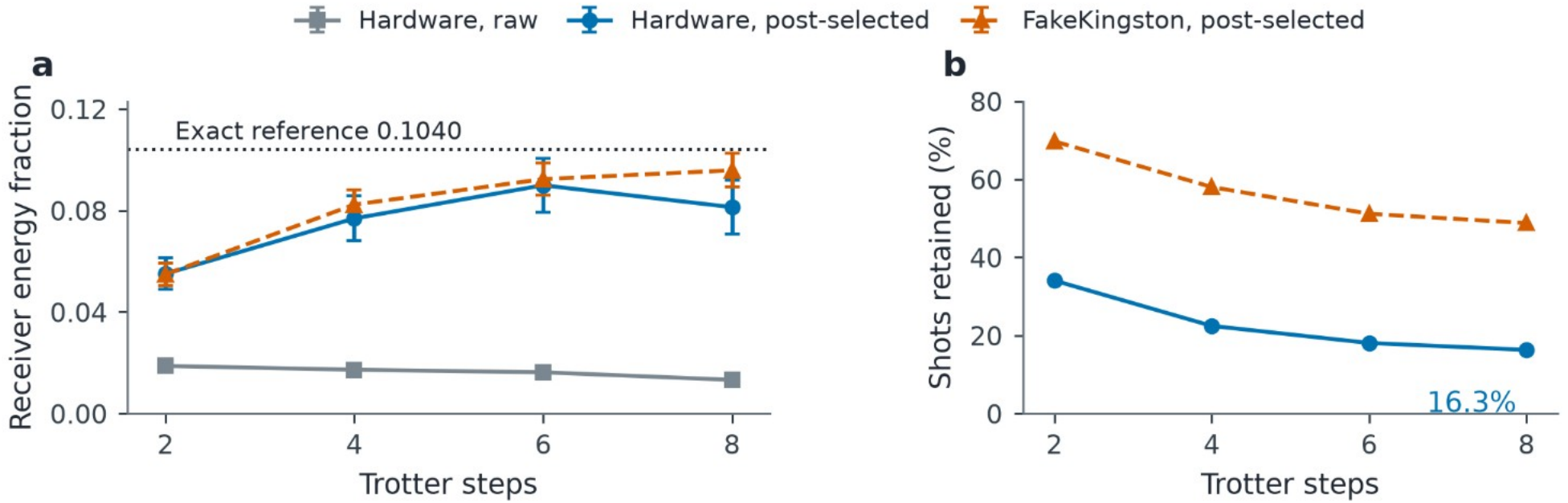


Simulator Validation



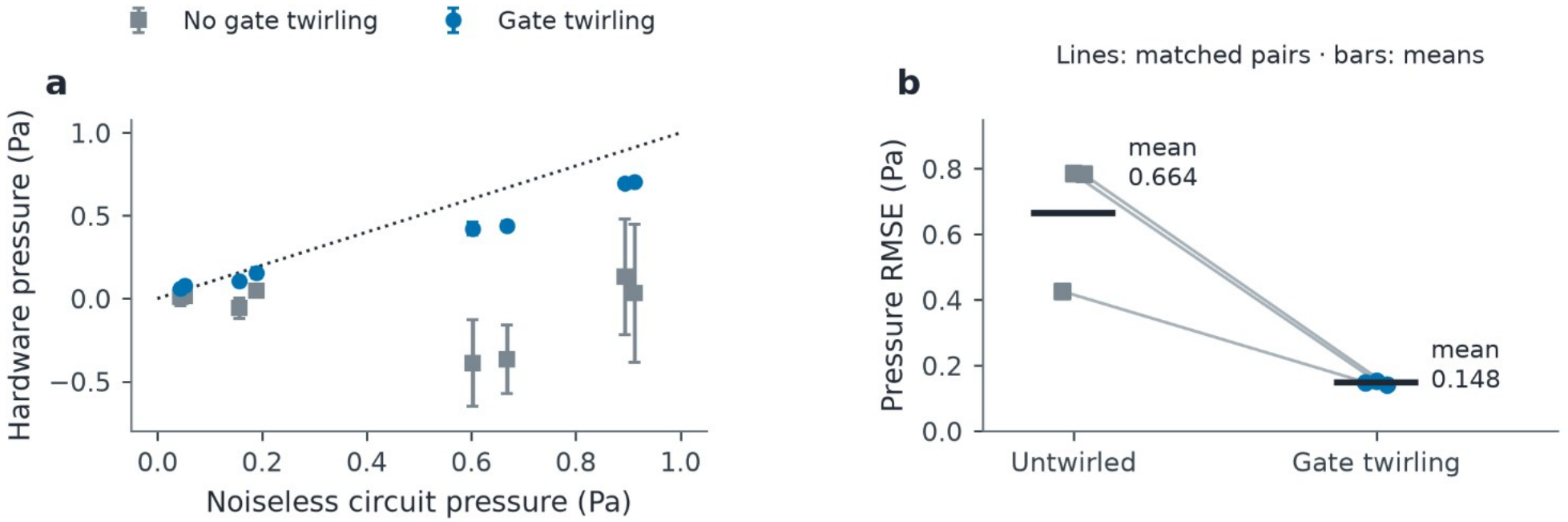
Ideal mean-trace error: 0.58%; with 0.1% gate noise: 8.85%.
All 3 ideal repeat errors exceeded 1%; this is a small synthetic pilot.

IBM Hardware: Wave-Energy Demo



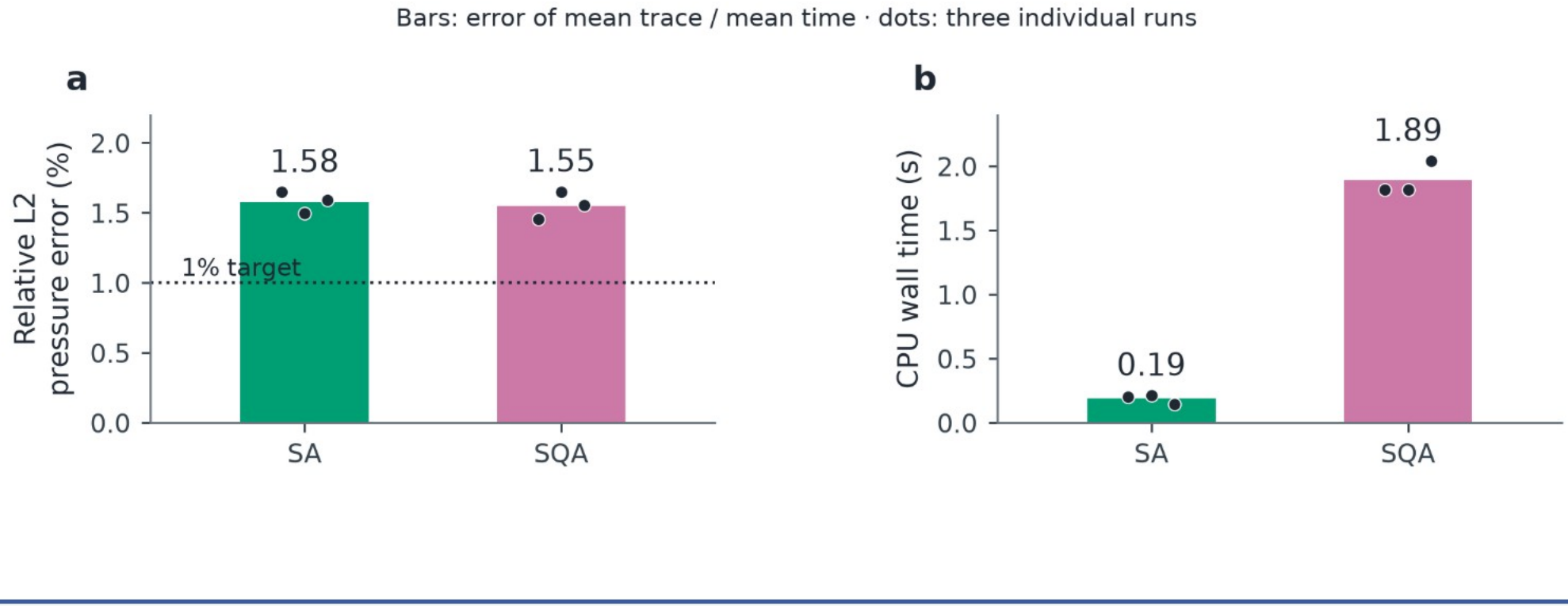
Post-selection recovered signal, but only 16.3% of shots survived at 8 steps.
15-qubit IBM test: one job, 4,000 shots/circuit; independent repeats pending.

IBM Hardware: Signed Pressure



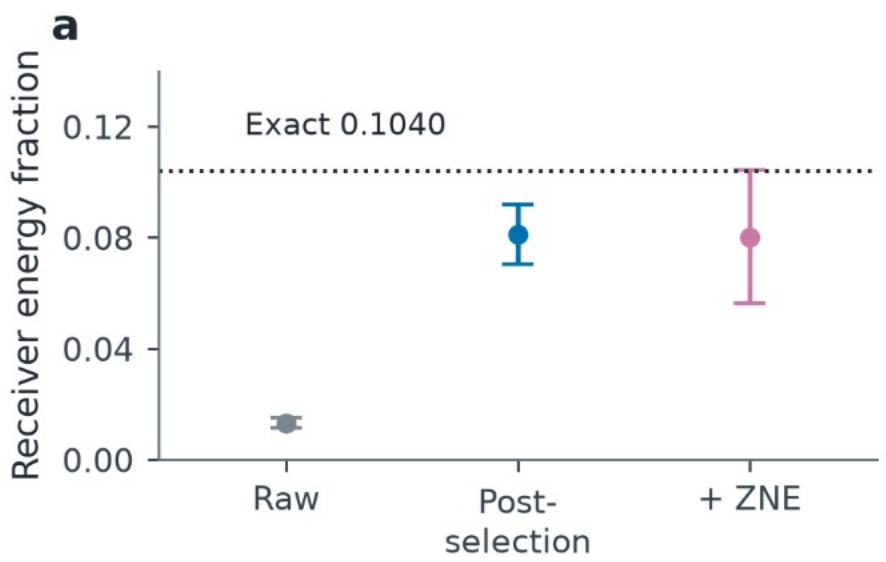
Gate twirling cut mean pressure error 78%, improving all 3 matched pairs.
6 IBM Kingston jobs; readout correction worsened all 6. Residual error remains.

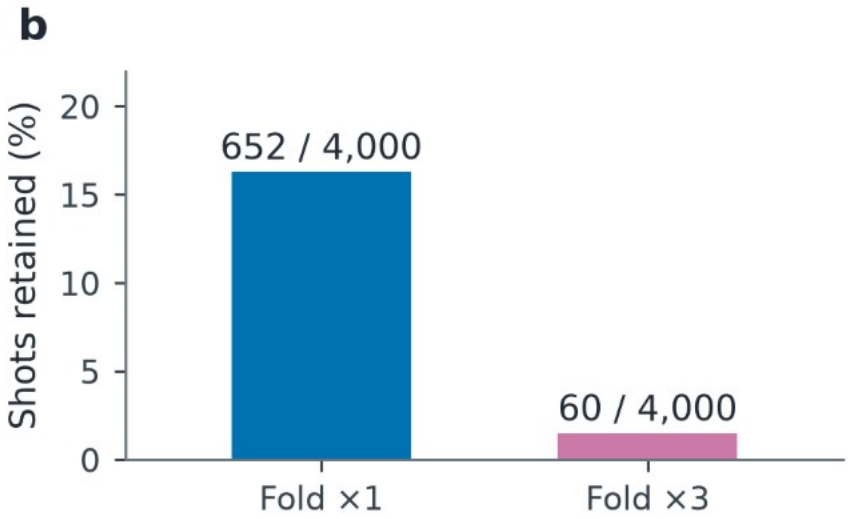
Simulated Quantum Annealing



Similar error: SA 1.58%, SQA 1.55%; SQA took ~10× longer on a CPU.
3 runs/method; neither met the every-repeat ≤1% target. No annealing hardware.

Quantum Error Mitigation

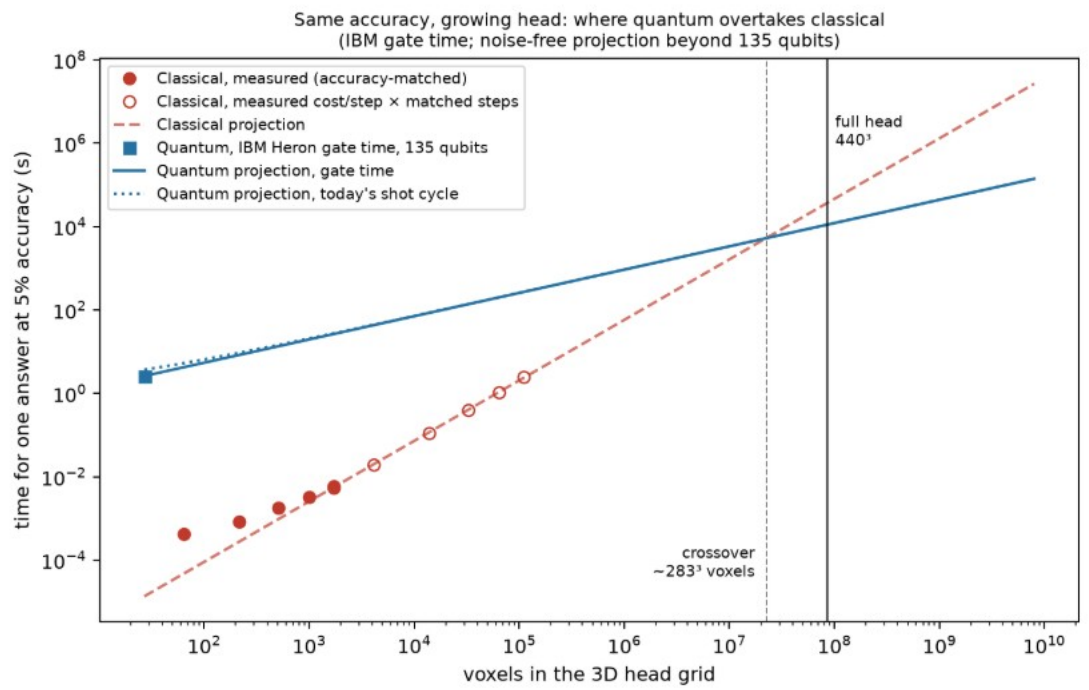




Post-selection cut absolute error ~75% for one observable in one job.
ZNE did not establish further improvement; only 60 folded shots survived.

Quantum vs Classical: Speed

Same accuracy, growing head: where quantum overtakes classical.



110× faster per circuit run: 6.3 µs on IBM's chip (gate + readout time) vs 695 µs classical, same problem

283³ voxels: crossover at matched 5% accuracy (projected)

~3.3× faster for a full 440³ head: ~3.1 h vs ~10.2 h, one wave simulation (projected)

CAVEATS

One run is one shot; an answer needs ~4,000 shots, so at today's size classical still wins the full answer. Measured to 135 qubits; beyond is a noise-free projection. Classical = our solver on one laptop core.

Team results §4b · scaling/crossover.py · 135-qubit circuit compiled to IBM Heron layout

Innovation & Creativity

What we built that's new: turning a 1D research method into a brain-imaging quantum pipeline.

01

Brain tissue drives the gates

Skull, brain and blood sound speeds set each coupling gate's strength, so the circuit carries a real head map.

Built on Schade 2024's 1D elastic method

02

From 1D to 2D and 3D

We extended the 1D algorithm to 2D and a 3D grid of 27 cells on 135 qubits.

3D circuit error 0.0057 vs exact

03

A physics-aware noise filter

Our encoding allows exactly one excitation. Any shot with more is noise, so we throw it out.

IBM hardware: 0.013 → 0.081 (exact 0.104)

04

Two encodings, two jobs

15-qubit unary circuit for wave energy; compact 4-qubit circuit for signed pressure.

Twirling cut pressure error 78% on IBM Kingston

05

One loop, four engines

Same physics run through classical, gate-based, annealing and IBM hardware for a fair head-to-head.

SQA matched SA accuracy, ~10× slower

06

Designed around the readout wall

Ask the circuit for one number, the misfit, instead of the whole wave.

Hybrid one-number readout prototyped

Team repos team-quantum-ultrasound, IBM_Qiskit_Hackathon · IBM jobs on ibm_marrakesh and IBM Kingston

Quantum Rationale & Understanding of Potential

As quantum computers grow and get cheaper, aperion grows with them.

TODAY

2026

Works, but no advantage yet

Small, noisy chips: we threw away 84% of shots to fix errors.

Every answer needs ~4,000 runs, paid by the second.

So for a 7-cell test, a laptop still wins.

NEXT

~2029

Error-corrected machines

IBM's roadmap: first fault-tolerant system, ~200 logical qubits.

Fewer errors means fewer wasted shots, so each answer gets cheaper.

FUTURE

2033+

Big enough for a head

Thousands of logical qubits on the roadmap.

Past ~283^3 voxels our method overtakes classical: ~3.3× for a full head, and the gap widens as grids grow.

We built and tested the software now, on real IBM hardware, so when the machines catch up, aperion is ready to run.

If it scales as planned, the same engine could image any part of the body, and a faster solver may let us use fewer ultrasound sensors.

IBM Quantum roadmap (Starling ~2029, Blue Jay 2033+) · crossover: team results §4b, noise-free projection

Conclusion · Lives saved

Bring the scan to the patient, and minutes turn into brains.

36 min

faster treatment when imaging comes to the patient

BEST-MSU, NEJM 2021

~68M

neurons saved per patient in those 36 minutes

36 × 1.9M, our calculation

49M

Americans live 60+ min from advanced stroke care

CDC 2025

HOW IT SCALES WITH QUANTUM

Today: 15 noisy qubits, a single line through the head. 2029: IBM Starling targets 200 error-corrected qubits. A whole head needs 29 of them and ~10^8+ reliable operations, asked for one number at a time.

WHAT MUST BE TRUE

Error-corrected qubits · efficient loading of the head · a one-number readout. Best published gain: polynomial, at best quartic in 3D (Bösch 2025).

BEST-MSU, NEJM 2021 · Saver 2006 · CDC 2025 · IBM Quantum roadmap · Bösch et al. 2025

Lessons Learned & Path to Scale

Evidence today, the remaining limits, and what we need to test next.

Completed evidence	Current limit	Next evidence needed
Gate twirling lowers pressure error in all three hardware pairs	Four-qubit synthetic model; residual error and drift remain	Repeat across calibrations; reduce gate depth
Ideal pressure simulation tracks the small reference	Noise and finite shots disturb reconstruction	Test stable gradients; complete a longer reconstruction
SQA solves the forward wave QUBO	No demonstrated gain over classical simulated annealing	Compare methods at matched accuracy and total cost

Future advantage requires efficient state preparation, sparse circuits and limited readout.

No end-to-end imaging speedup, clinical validation or reliable bleed detection has been demonstrated.

Next: reduce residual gate error → stable longer reconstruction → larger matched tests.



Imaging beyond what's visible.

References

1. Guasch, Calderón Agudo, Tang, Nachev & Warner. Full-waveform inversion imaging of the human brain. npj Digital Medicine 3, 28 (2020). Figures CC BY 4.0.
2. Costa, Jordan & Ostrander. Quantum algorithm for simulating the wave equation. Phys. Rev. A 99, 012323 (2019).
3. Schade et al. A quantum computing concept for 1-D elastic wave simulation. arXiv:2312.14747 (2024) · Bösch et al. Phys. Rev. Research (2025).
4. Cueto et al. Stride (2022) · Plessix 2006 · Brassard, Høyer, Mosca & Tapp 2002.
5. AHA Heart Disease & Stroke Statistics 2026 · 2026 AHA/ASA guideline · Saver, Stroke 2006 · CDC 2025 · BEST-MSU, NEJM 2021.
6. Kadowaki & Nishimori 1998 · Farhi 2014 · Peruzzo 2014 · Bowles, Ahmed & Schuld 2024 · HHL 2009 · Aaronson 2015 · Lloyd 1996.
7. Brain anatomy: MNI152 ICBM 2009 template (McConnell Brain Imaging Centre). Team code: team-quantum-ultrasound repo.

Backup slide, not presented

Appendixes

The same sound-and-math recipe can image almost anything.

Stroke in the ambulance

Clot or bleed before the hospital

Rural clinics

No CT nearby, a helmet and the cloud

Breast and limb

Ultrasound tomography is already FDA-approved for breast

Beyond medicine

Seismic imaging, non-destructive testing

aperion

We believe the wall is compute, not physics. Classical hardware will get there first; quantum is how the same scan could reach every clinic on the cloud. Every minute we save is 1.9 million neurons.

Results & Validation

Everything we found, on one slide.

ACCURACY

0.58%
pressure error, ideal quantum circuit vs exact

0.993
match between our solver and the paper group's software

1,576
m/s bleed recovered by our FWI (true 1,578)

NOISE CONTROL

−78%
pressure error with gate twirling on IBM Kingston

~75%
energy error removed by post-selection

8.85%
error once 0.1% gate noise is added

REAL IBM HARDWARE

0.081
energy measured on ibm_marrakesh after filtering (exact 0.104)

24,000
shots in one job · 9 s of quantum time

16.3%
of shots survive filtering at 8 steps

SPEED

110×
faster per circuit run: 6.3 µs vs 695 µs

283^3
voxels: where quantum overtakes classical at 5% accuracy*

~3.3×
faster for a full head: ~3.1 h vs ~10.2 h*

Also: 3D circuit on 135 qubits, error 0.0057 · simulated quantum annealing matched classical accuracy but ran ~10× slower · *projections assume noise-free hardware beyond 135 qubits.

Team results · IBM jobs on ibm_marrakesh and IBM Kingston · scaling/crossover.py



Quantum Rationale & Understanding of Potential

How the image is found: guess, measure, step downhill.

1. Residual: predicted minus measured

$$\mathbf{r}(\mathbf{c}) = F(\mathbf{c}) - \mathbf{d}$$

2. Squash every mismatch into one number

$$E(\mathbf{c}) = \tfrac{1}{2}\,\|\mathbf{r}(\mathbf{c})\|_2^2 = \tfrac{1}{2}\sum_{s,r,n}\big(F_{s,r,n}(\mathbf{c}) - d_{s,r,n}\big)^2$$

3. Chain rule gives the slope per voxel (adjoint)

$$\frac{\partial E}{\partial c_j} = \sum_{\ell} r_\ell\,\frac{\partial F_\ell}{\partial c_j} \quad\Longrightarrow\quad \nabla E(\mathbf{c}) = J_F(\mathbf{c})^{\top}\,\mathbf{r}(\mathbf{c})$$

4. Step downhill, keep speeds physical, repeat

$$\mathbf{c}_{k+1} = \Pi_{\mathcal{C}}\Big[\mathbf{c}_k - \eta_k\big(\nabla E(\mathbf{c}_k) + \lambda\,\nabla R(\mathbf{c}_k)\big)\Big]$$

5. Why it took 32 hours

$$\text{cost} \approx O(I \cdot S \cdot N_t \cdot V) \approx 80 \times 100 \times 2 \times 2400 \times 10^8 \approx 4 \times 10^{15}\ \text{updates}$$

Plessix 2006 · Stride, Cueto et al. 2022 · Cost line: our estimate from Guasch et al.

Quantum Rationale & Understanding of Potential

How a computer simulates sound, step by step.

1. Start from the wave equation (1D for clarity)

$$\frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} = \frac{\partial^2 p}{\partial x^2} + q$$

2. Cut space into cells h and time into ticks Δt

$$x_j = j\,h, \qquad t_n = n\,\Delta t, \qquad p_j^n \approx p(x_j, t_n)$$

3. Replace each curvature with neighbour differences

$$\frac{\partial^2 p}{\partial x^2} \approx \frac{p_{j+1}^n - 2p_j^n + p_{j-1}^n}{h^2}, \qquad \frac{\partial^2 p}{\partial t^2} \approx \frac{p_j^{n+1} - 2p_j^n + p_j^{n-1}}{\Delta t^2}$$

4. Solve for the next tick: the update rule

$$p_j^{n+1} = 2p_j^n - p_j^{n-1} + \frac{c_j^2\,\Delta t^2}{h^2}\left(p_{j+1}^n - 2p_j^n + p_{j-1}^n\right) + c_j^2\,\Delta t^2\,q_j^n$$

5. Keep the time step small enough to stay stable

$$\frac{c_{\max}\,\Delta t}{h} \le 1 \ \ (1\mathrm{D}), \qquad \frac{c_{\max}\,\Delta t}{h} \le \frac{1}{\sqrt{3}} \ \ (3\mathrm{D})$$

Team math notes · Guasch et al. 2020 · 3D uses six neighbours

Quantum Rationale & Understanding of Potential

Why quantum: sound becomes a Schrödinger equation.

1. The grid's wave equation (B = neighbour map)

$$\frac{d^2\boldsymbol{\phi}}{dt^2} = -B\,B^\dagger\,\boldsymbol{\phi}$$

2. Stack velocity and stretch into one vector ψ

$$\boldsymbol{\psi} = \begin{pmatrix} \dot{\boldsymbol{\phi}} \\ -i\,B^\dagger\boldsymbol{\phi} \end{pmatrix}$$

3. It obeys Schrödinger's equation

$$i\,\frac{d\boldsymbol{\psi}}{dt} = H\,\boldsymbol{\psi}, \qquad H = \begin{pmatrix} 0 & B \\ B^\dagger & 0 \end{pmatrix}$$

4. Wave energy = probability: both conserved

$$\|\boldsymbol{\psi}\|^2 = \|\dot{\boldsymbol{\phi}}\|^2 + \|B^\dagger\boldsymbol{\phi}\|^2 = \text{wave energy (conserved)}$$

5. Run it as a circuit: Trotter steps

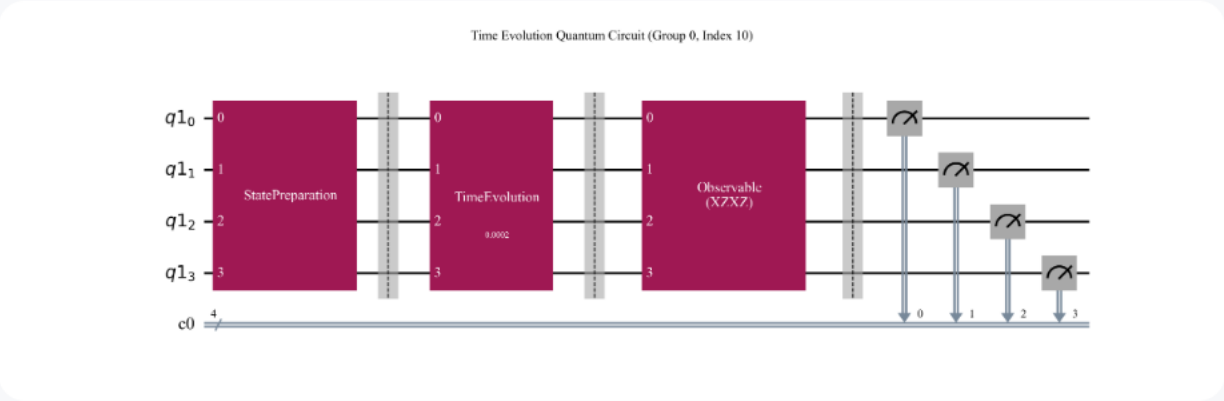
$$\boldsymbol{\psi}(t) = e^{-iHt}\,\boldsymbol{\psi}(0) \approx \left(e^{-iH_{\text{even}}\Delta t}\,e^{-iH_{\text{odd}}\Delta t}\right)^n \boldsymbol{\psi}(0)$$

Costa, Jordan & Ostrander 2019 (H carries 1/a) · Schade et al. 2024 · Gain: polynomial, at best quartic in 3D (Bösch 2025)

Quantum Rationale & Understanding of Potential

The 1D quantum algorithm we build on: how a wave runs on qubits.

1. **Discretize the wave**
   1D elastic wave on a grid, with varying density ρ and stiffness μ
2. **Rewrite as Schrödinger**
   Velocity and strain stack into $|\psi\rangle$; $i\partial_t|\psi\rangle = H|\psi\rangle$
3. **Load the wave**
   2^n grid points live in just n qubits
4. **Evolve**
   Apply e^(−iHt): the wave moves inside the qubits
5. **Measure**
   Repeat and count to read the wavefield back



64 grid points → 6 qubits. A whole head's 341M sites → 29.

Schade, Bösch, Hapla & Fichtner, arXiv:2312.14747v2 (2024), Appendix figure, CC BY 4.0

Problem Definition & Relevance

Competitive advantage: the only option that goes to the patient with no radiation and no magnet.

Category	aperion helmet	Hospital CT	Hospital MRI	CT ambulance	Portable MRI
Goes to the patient	✓	✗	✗	✓	✓
No ionizing radiation	✓	✗	✓	✗	✓
No magnet, no shielding	✓	✓	✗	✓	✗
Sees through the skull	✓	✓	✓	✓	✓
Image in minutes	✗ 32 h today	✓	✓	✓	✓
Proven in patients	✗ simulation	✓	✓	✓	✓

Its two gaps are time and clinical proof. Time is the one we attack.

RadiologyInfo · BEST-MSU, NEJM 2021 · Hyperfine Swoop, Nat Commun 2021 · Guasch et al. 2020